Learning rate experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make controls and gradient scale visible.

Use exact recurrence and complete curves;separate crossings,unchanged oscillation and growing loss with model-specific bounds.

Keep one habit: Assess the new objective's two-update paths and bounded conclusion using checked keys.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Run a **controlled comparison** of **learning rate**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

An epoch visits the data once, but each batch causes an update. More batches mean more updates under the stated rule.

Opening retrieval

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish an example a batch an update and an epoch

Reference example, not the only acceptable wording: Teacher explanation: Batch lengths 4, 4, 2; three updates per epoch, nine total updates and 30 example visits.

Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a **learning rate** to an ordered gradient

Reference example, not the only acceptable wording: Teacher explanation: New a = 2 - 0.2 times -3 = 2.6.

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate per-example squared error and mean squared error

Reference example, not the only acceptable wording: Teacher explanation: Residuals are -1 and 2, squares 1 and 4, MSE 2.5.

Pictured end goal

By the end, I can explain **learning rate** experiments using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Batching and epochs

Updating parameters

Loss functions

For the compact alternative schedule, use two 45-minute sessions and connect signed distance to squared loss before introducing the factor formula. This is a fully explained quadratic experiment with a frozen hidden representation.

Avoid general prescriptions about rate values. Students should distinguish an oscillating parameter from increasing loss.

Outcomes

Run a **controlled comparison** of **learning rate**s

Explain slow progress **oscillation** and increasing loss in a toy model

Report the limits of a **training** experiment

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach controlled paths rather than learning-rate folklore

> Keep the three quadratic objectives and their gradient scales distinct.

Main experiment,complete trajectories and guided key

Use the same restricted network h = ReLU(x), y = vh, with only v trained. Keep the fictional **training** pairs (1, 2) and (2, 4), start v = 0, and use one full-data batch per update.

Compare rates 0.05, 0.2, 0.3, 0.4 and 0.5 for exactly four updates each. Restart at zero for every run.

The **training** objective is ordinary MSE: L(v) = 2.5 times (v - 2) squared. Its gradient is 5 times (v - 2). A step is new v = v - rate times 5 times (v - 2).

This small quadratic lets us explain each result rather than treating the rate as a magic setting.

The same rate need not behave this way for another model, **input** scale, loss convention or batch rule. Rate 0.2 is an exact one-step solution here because of this particular gradient formula.

Write d = v - 2, the signed distance from this objective minimum. Substituting the update gives new d = (1 - 5 times rate) times old d. At rate 0.05 the factor is 0.75: distance shrinks without changing sign.

At rate 0.3 it is -0.5: distance changes sign and halves. Both can reduce loss.

The helper assumes a finite rate and nonnegative integer update count. Its five logged states include the initial value.

At 0.3 the parameter sequence is [0, 3, 1.5, 2.25, 1.875] and loss is [10, 2.5, 0.625, 0.15625, 0.0390625]. Crossing the minimum does not itself mean **training** failed.

Rate	All five parameter states(0–4 updates)	All five training MSEs	Distance multiplier 1–5 rate
0.05	[0.0, 0.5, 0.875, 1.15625, 1.3671875]	[10.0, 5.625, 3.1640625, 1.77978515625, 1.001129150390625]	0.75
0.2	[0.0, 2.0, 2.0, 2.0, 2.0]	[10.0, 0.0, 0.0, 0.0, 0.0]	0.0
0.3	[0.0, 3.0, 1.5, 2.25, 1.875]	[10.0, 2.5, 0.625, 0.15625, 0.0390625]	-0.5
0.4	[0.0, 4.0, 0.0, 4.0, 0.0]	[10.0, 10.0, 10.0, 10.0, 10.0]	-1.0
0.5	[0.0, 5.0, -2.5, 8.75, -8.125]	[10.0, 22.5, 50.625, 113.90625, 256.2890625]	-1.5

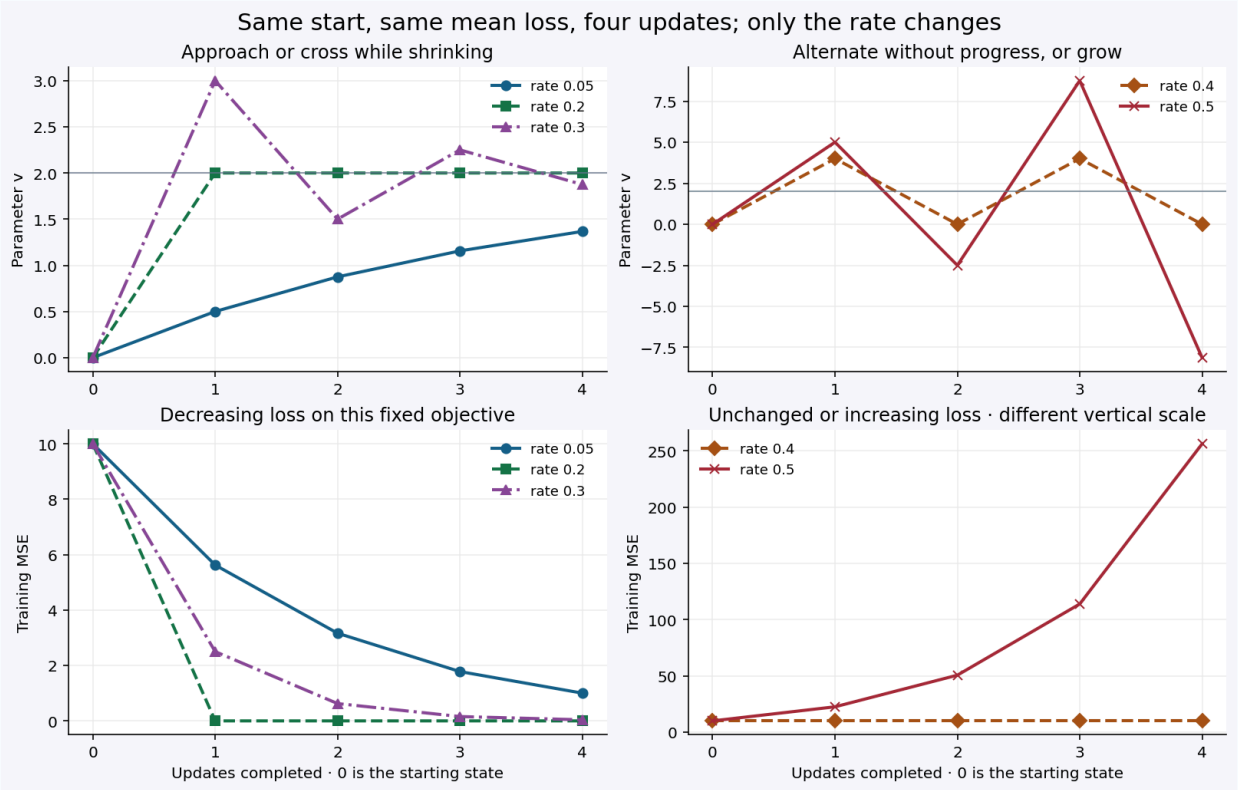
The saved curves are computed from the exact declared update v←v−rate×5(v−2), starting 0 and logging states 0 through4. They are reproducible toy calculations, not measurements from a real neural training service. Write d=v−2:each step multiplies d by1−5 rate and multiplies loss by its square. Crossing sides can still shrink the distance.

The numerical helper assumes finite rate and a nonnegative integer update count; it does not validate them or choose a rate for an unseen task.

Guided: multiplier -1; v becomes 4 then 0, with MSE 10 at both states. Parameter change does not guarantee decreasing loss.

Exit: for the fixed two-example objective, start and four-update budget, rate 0.2 reached zero **training** MSE after one update. This does not demonstrate an optimal rate or generalisation for other networks or data.

Labelled parameter/loss curves



Separate pictured two-update experiment

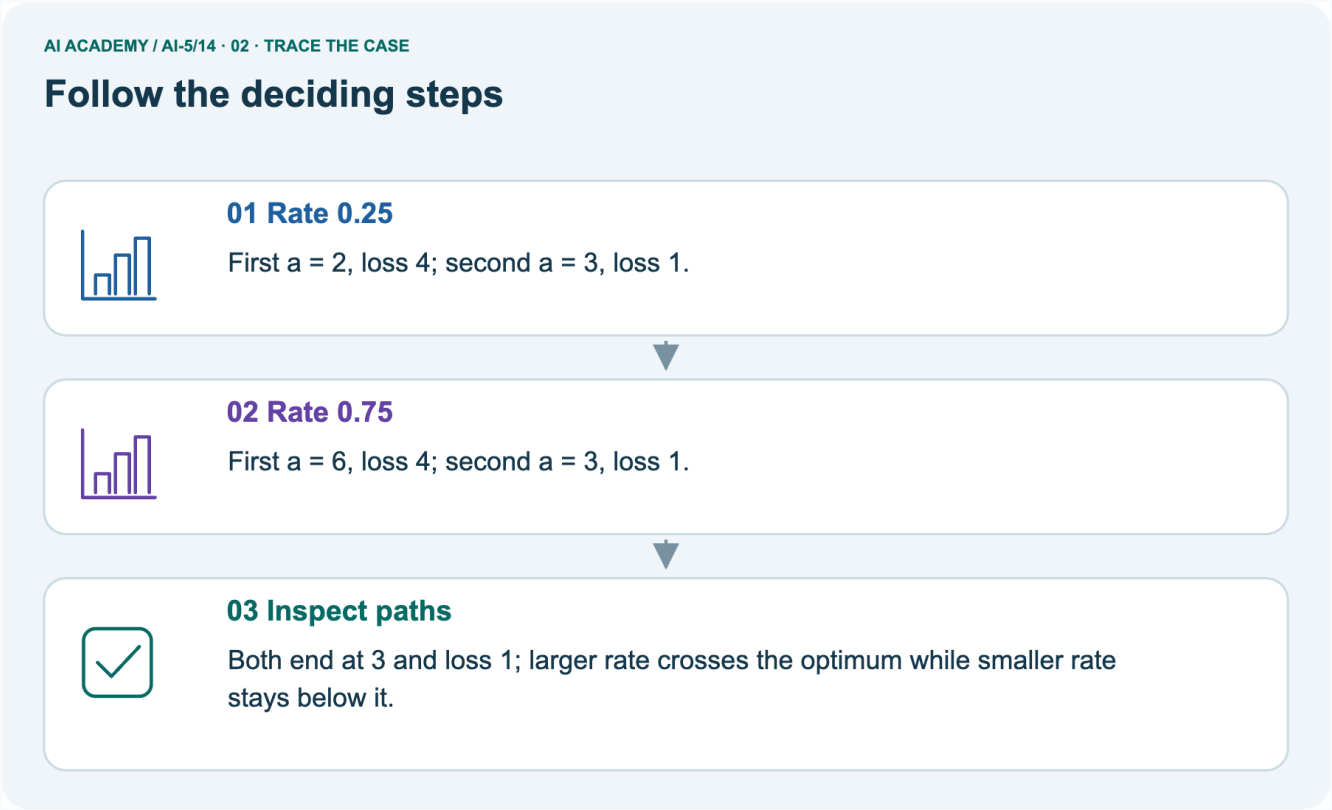
Compare **learning rate**s with identical starting loss and update counts.

Loss $J(a)=(a-4)^2$; gradient $2(a-4)$; start $a = 0$. Compare two updates at rates 0.25 and 0.75.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses a different objective $J(a)=(a-4)^2$, gradient $2(a-4)$, start 0 and two updates. Rate.25 gives a2 then3, loss 4 then1; rate.75 gives a6 then3, also loss 4 then1. Different paths happen to share the final result. A separate restarted rate 1 run gives a8 then0, loss 16 both times. These rates belong to that objective, not the main $2.5(v-2)^2$. The independent task supplies a third objective centred at1.

Trace crossing,matching endpoints and unchanged loss



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Rate 0.25	First a = 2, loss 4; second a = 3, loss 1.
2. Rate 0.75	First a = 6, loss 4; second a = 3, loss 1.
3. Inspect paths	Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Keep the starting parameter and update budget identical to compare rates fairly. Rate one-quarter approaches the optimum from below; rate three-quarters crosses it and comes back.

Their final losses happen to match after two updates, despite different paths. Rate one repeatedly overshoots by the same distance, so using the correct gradient alone does not ensure improvement.

These quadratic paths explain step size. Real nonconvex networks can have more complicated rate behaviour.

Change: Add a rate 1 run from a = 0.

Prediction: Its path is 8 then 0, with loss 16 at both steps.

Reason: The update alternates across 4 without reducing the initial loss.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Learning rate	The multiplier controlling the size of a gradient update
Controlled comparison	An experiment changing a named factor while holding other stated conditions fixed
Oscillation	Repeated movement from one side of a value to the other
Divergence	Movement away from a target solution with growing error in this experiment

The update factor explains three behaviours

RATE AND FACTOR	SIGNED DISTANCE BEHAVIOUR	TRAINING LOSS BEHAVIOUR
0.05 gives 0.75	Same side and shrinking	Multiplied by 0.5625
0.3 gives -0.5	Alternating and shrinking	Multiplied by 0.25
0.5 gives -1.5	Alternating and growing	Multiplied by 2.25

For this objective, loss is 2.5 times squared distance. Therefore the loss multiplier is the square of the distance multiplier.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

The update factor explains three behaviours

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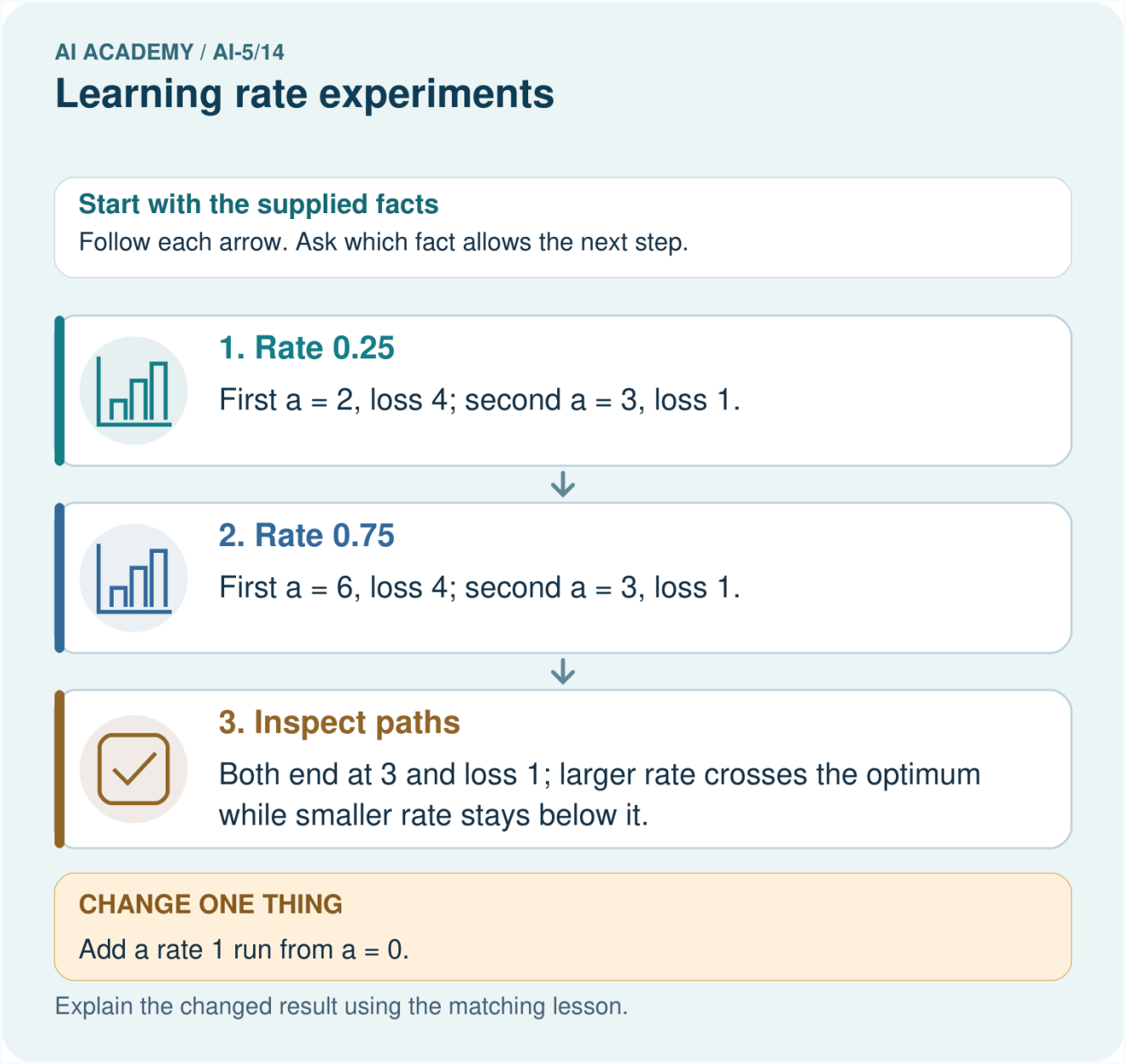
For rate 0.4, calculate its distance multiplier, the first two new v values and both losses. Does a changing parameter guarantee progress?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Learning rate experiments

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: At rate 0.3 the first update gives v = 3, past the minimum at 2. Calculate the next v and loss. Has the first crossing proved **divergence**?

Teacher check: Teacher explanation: Next gradient 5 gives v = 1.5 and loss 0.625. No. Signed distance changes side but shrinks by half each update; loss decreases.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Learning rate** experiments. Proposed explanation: “Crossing the minimum means **divergence**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Compare distance magnitudes or losses; alternating sides can converge.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Use the same objective, start a = 0 and two-update budget; report full paths, not only the preferred run's endpoint. Changed-case check: Its path is 8 then 0, with loss 16 at both steps.

The update alternates across 4 without reducing the initial loss.

Changed-case reasoning: Its path is 8 then 0, with loss 16 at both steps. The update alternates across 4 without reducing the initial loss.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a number line with the minimum marked and a table for old value, gradient, new value and loss.

Extension: derive 0 < rate < 1 for strict loss decrease on J(a) = (a - 1) squared away from the minimum, and compare with the lesson bound 0.4.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve epoch and update counts; agree on one full-data batch and four updates per run.
7 to 18 minutes	Calculate the first row of each rate run and keep all resets visible.
18 to 32 minutes	Derive new d from the update rule using substitution; complete W1.
32 to 45 minutes	Run or trace Python and compare rate 0.3 with W2, marking sign and magnitude separately.
Session 2 start to 8 minutes	Complete the guided rate 0.4 example and explain why changing values can leave loss unchanged.
8 to 20 minutes	Use W3 to connect the loss scale with the gradient and an equivalent rate.
20 to 35 minutes	Collect the three independent W4 traces with a fresh reset before each.
35 to 45 minutes	Use W5 and the exit response, then administer the fresh retry and record the experimental limits.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Mainrate.3 secondstate1.5/loss.625, crossingnotdivergence; rate.4v4/0 loss10/10. W3SSEequivalentrate.15. W5 restart same original state and hold batch/objective/budget fixed before attributing change to rate. Independentstart0:rate.25a.5/.75 loss.25/.0625; rate.5a1/1 loss0/0; rate 1a2/0 loss1/1. Rate.5 lowestobjective aftertwo updates in this supplied comparison, not all networks.

Curve JSON and code preserve exact data.

Diagnose the current misconceptions

Crossing the minimum means **divergence**: Compare distance magnitudes or losses; alternating sides can converge.

A larger rate is always faster: The tested 0.5 rate increases this objective; 0.4 keeps it unchanged.

The best rate is independent of loss scaling: Scaling the loss scales its gradient and changes the step at a fixed rate.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Use J(a) = (a - 1) squared with gradient 2(a - 1), start a = 0 and two updates. Restart for each rate 0.25, 0.5 and 1. Compute the two new a values and losses for each run.

Identify the result supported after two updates.

Teacher explanation: At 0.25: a = 0.5 then 0.75; losses 0.25 then 0.0625. At 0.5: a = 1 then 1; losses 0 then 0. At 1: a = 2 then 0; losses 1 then 1. Rate 0.5 attains the lowest objective in this comparison.

The useful rate differs from the lesson model.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace slow progress

For rate 0.05, start v = 0 and compute two updates and their **training** MSE values using the lesson objective.

Teacher explanation: First v = 0.5 with MSE 5.625. Second gradient -7.5 gives v = 0.875 and MSE 3.1640625. Distance is multiplied by 0.75 each step.

W2 Explain a crossing

At rate 0.3 the first update gives v = 3, past the minimum at 2. Calculate the next v and loss. Has the first crossing proved **divergence**?

Teacher explanation: Next gradient 5 gives v = 1.5 and loss 0.625. No. Signed distance changes side but shrinks by half each update; loss decreases.

W3 Preserve an update after rescaling loss

Replace MSE by the sum of squared errors on the two **training** pairs. Which rate reproduces the MSE run at rate 0.3? Explain using gradients.

Teacher explanation: The sum gradient is twice the mean gradient, so rate 0.15 reproduces the original steps. Rate 0.3 would double each step from the same state.

W5 Find an unfair comparison

Run A starts v = 0 with full-data batches for four updates. Run B starts at the final A value, changes both rate and batch size, then compares final **training** losses. Can the difference be attributed only to rate?

Propose a repair.

Teacher explanation: No. Initial state and batch rule also changed. Restart every run at the same v, hold data, loss, batch rule and update budget fixed, and vary only rate.

Report the traces and separate any later experiment with a different factor.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Guided and exit questions

Guided: multiplier -1; v becomes 4 then 0, with MSE 10 at both states. Parameter change does not guarantee decreasing loss.

Exit: for the fixed two-example objective, start and four-update budget, rate 0.2 reached zero **training** MSE after one update. This does not demonstrate an optimal rate or generalisation for other networks or data.

W1 Trace slow progress

W2 Explain a crossing

W3 Preserve an update after rescaling loss

W4 Compare a new objective independently

Use J(a) = (a - 1) squared with gradient 2(a - 1), start a = 0 and two updates. Restart for each rate 0.25, 0.5 and 1. Compute the two new a values and losses for each run. Identify the result supported after two updates.
</details>

Teacher explanation: At 0.25: a = 0.5 then 0.75; losses 0.25 then 0.0625. At 0.5: a = 1 then 1; losses 0 then 0. At 1: a = 2 then 0; losses 1 then 1. Rate 0.5 attains the lowest objective in this comparison.

The useful rate differs from the lesson model.

W5 Find an unfair comparison

Independent assessment and fresh retry

W4 must include all six new states and losses with no carry-over between runs. Fresh retry: J(c) = 2(c - 3) squared, gradient 4(c - 3), start c = 1. At rate 0.125, c becomes 2 then 2.5 with losses 2 then 0.5.

At rate 0.25, c becomes 3 then 3 with losses 0 then 0. Ask for a conclusion limited to these runs.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **learning rate** experiments **evidence** sheet to Image-pattern Network Investigation. Show how you can run a **controlled comparison** of **learning rate**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide a number line with the minimum marked and a table for old value, gradient, new value and loss.

Extension: derive 0 < rate < 1 for strict loss decrease on J(a) = (a - 1) squared away from the minimum, and compare with the lesson bound 0.4.

Week 15 separates **training** and validation curves.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/15, ai-5/17, ai-5/20. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Run a **controlled comparison** of **learning rate**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: At rate 0.3 the first update gives v = 3, past the minimum at 2. Calculate the next v and loss. Has the first crossing proved **divergence**?

Reasoned practice answer: Teacher explanation: Next gradient 5 gives v = 1.5 and loss 0.625. No. Signed distance changes side but shrinks by half each update; loss decreases.

Check every learner before discussion: Use the worked case for **Learning rate** experiments. Proposed explanation: “Crossing the minimum means **divergence**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Compare distance magnitudes or losses; alternating sides can converge.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add a rate 1 run from a = 0. Require a **prediction** and a reason before comparing. Expected result: Its path is 8 then 0, with loss 16 at both steps.

Reason: The update alternates across 4 without reducing the initial loss.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

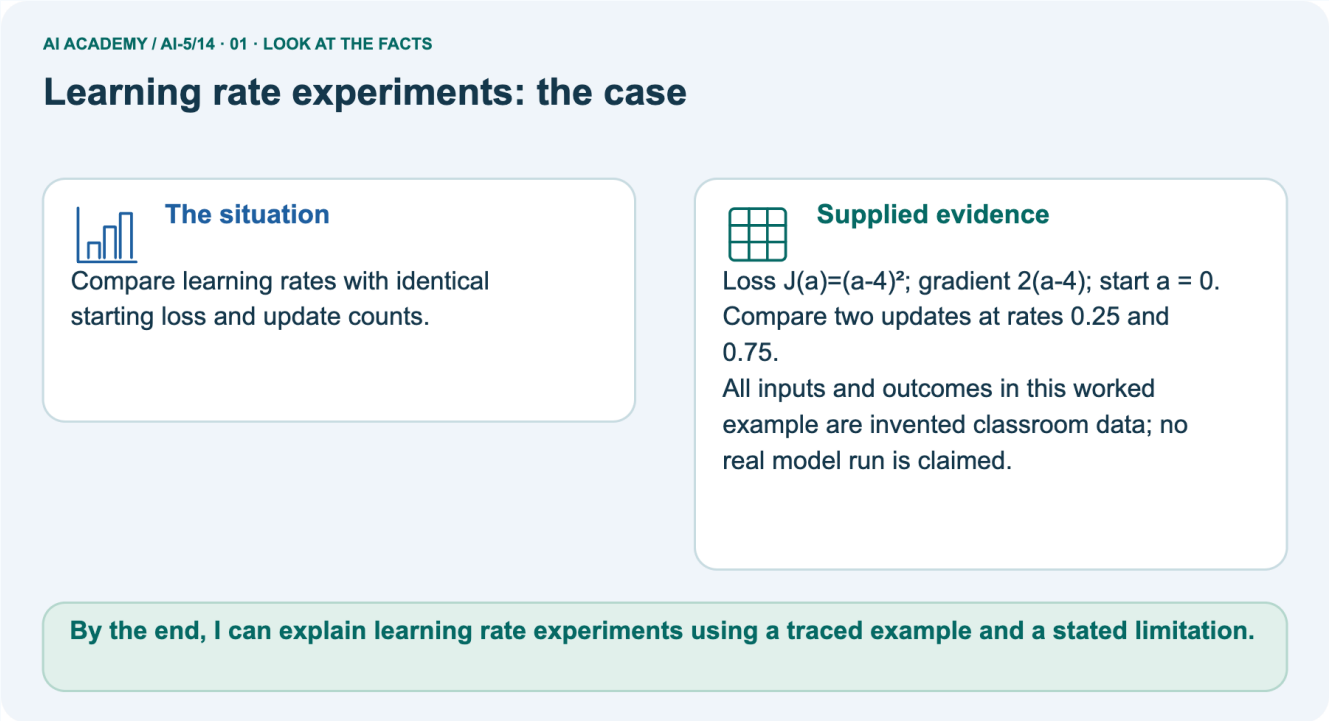
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Learning rate experiments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

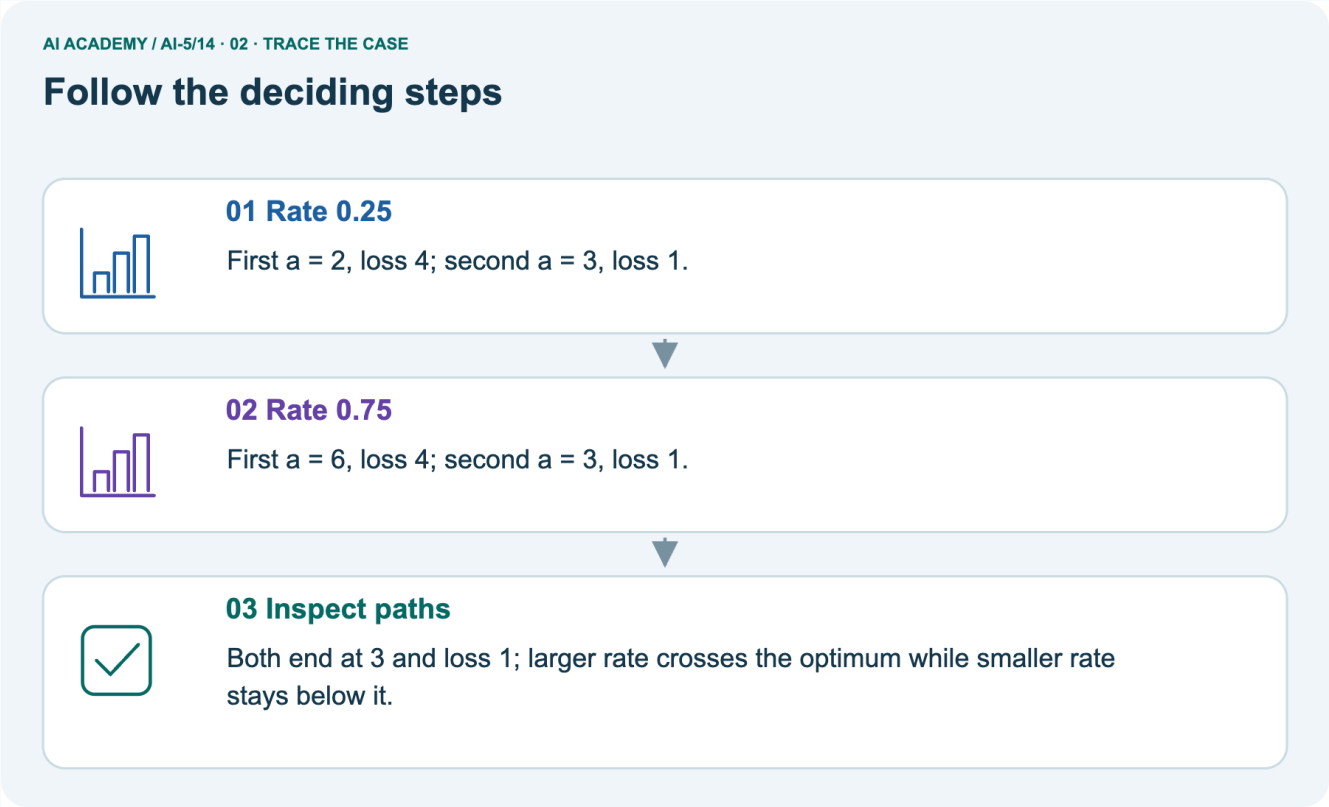
Compare **learning rate**s with identical starting loss and update counts. Supplied facts: Loss J(a)=(a-4)²; gradient 2(a-4); start a = 0. Compare two updates at rates 0.25 and 0.75.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



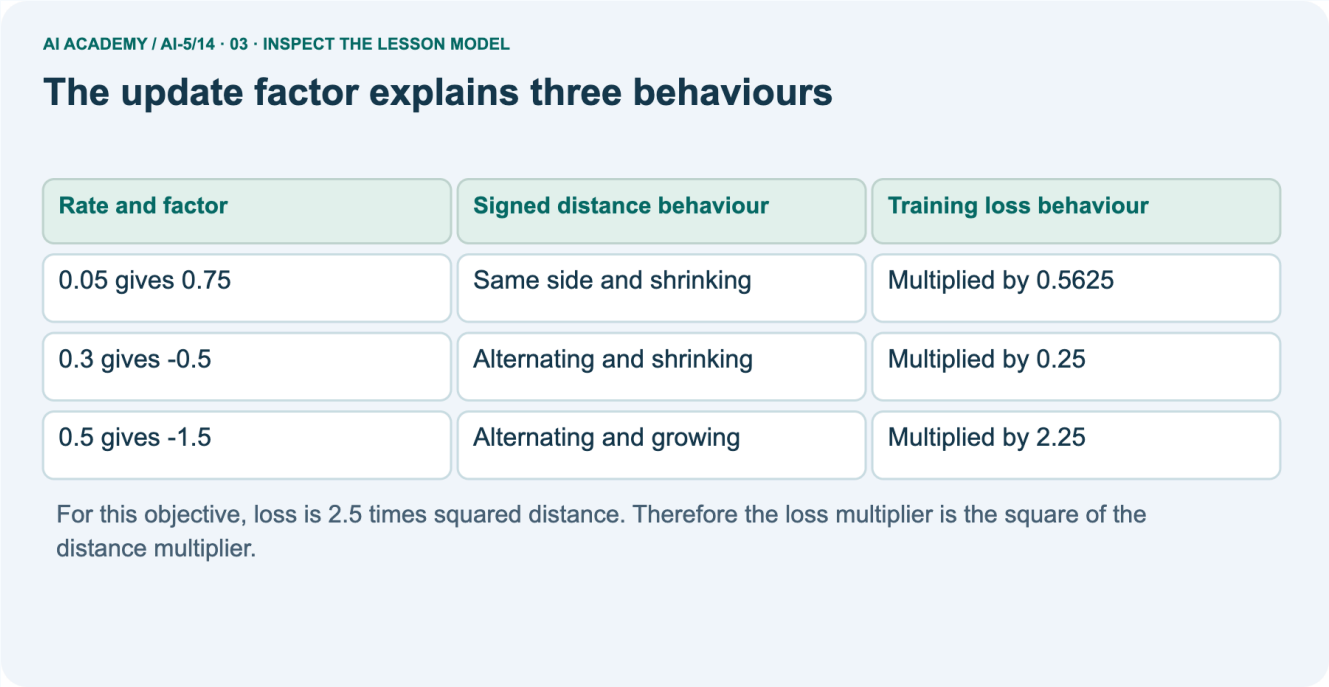
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Rate 0.25: First a = 2, loss 4; second a = 3, loss 1. → Rate 0.75: First a = 6, loss 4; second a = 3, loss 1. → Inspect paths: Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. The update factor explains three behaviours



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The update factor explains three behaviours

Rate and factor	Signed distance behaviour	Training loss behaviour
0.05 gives 0.75	Same side and shrinking	Multiplied by 0.5625
0.3 gives -0.5	Alternating and shrinking	Multiplied by 0.25
0.5 gives -1.5	Alternating and growing	Multiplied by 2.25

For this objective, loss is 2.5 times squared distance. Therefore the loss multiplier is the square of the distance multiplier.

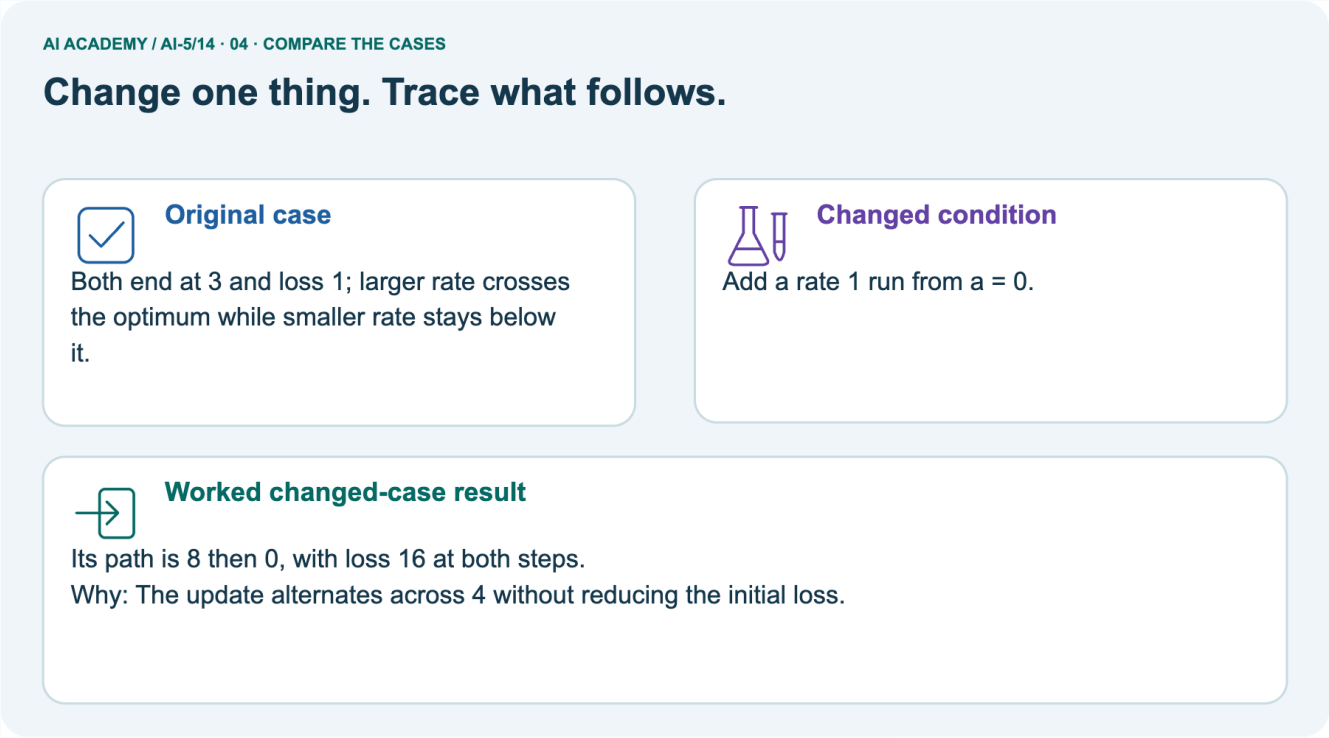
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The update factor explains three behaviours. 0.05 gives 0.75 | Same side and shrinking | Multiplied by 0.5625. 0.3 gives -0.5 | Alternating and shrinking | Multiplied by 0.25. 0.5 gives -1.5 | Alternating and growing | Multiplied by 2.25

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it. Changed condition: Add a rate 1 run from a = 0. Expected result: Its path is 8 then 0, with loss 16 at both steps.

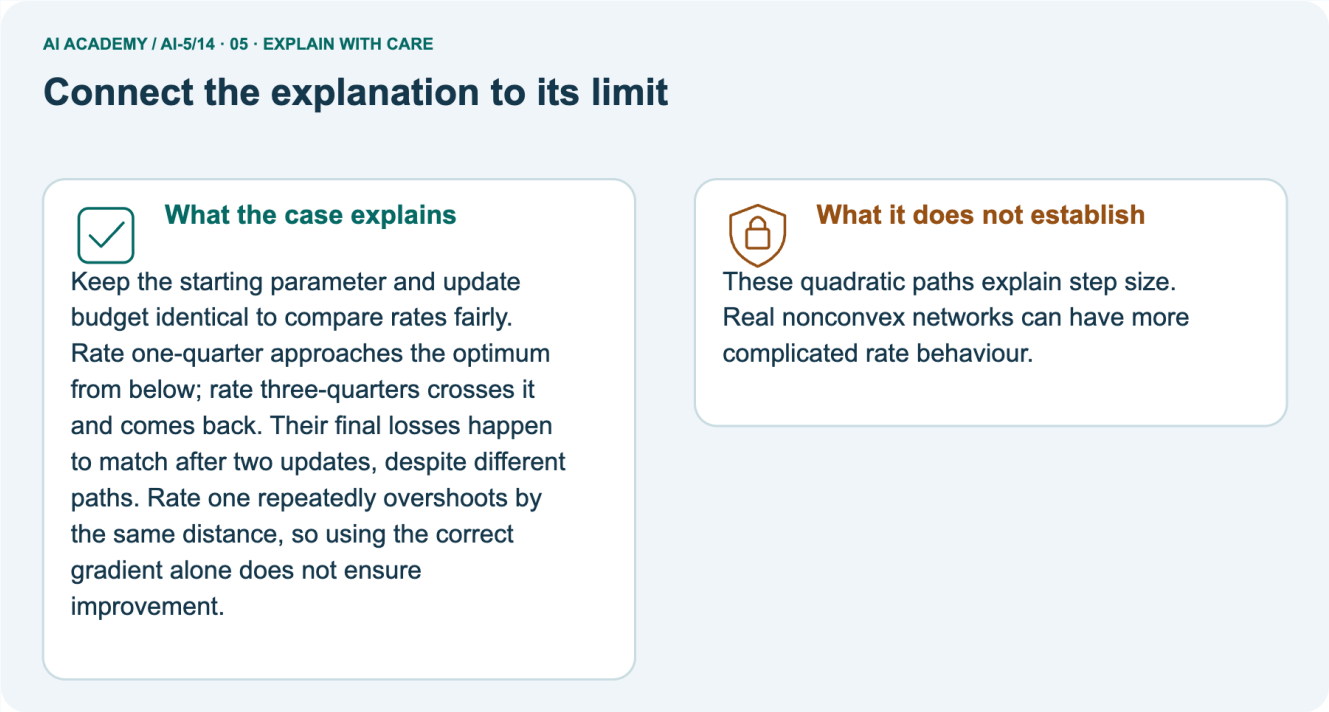
Reason: The update alternates across 4 without reducing the initial loss.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The update alternates across 4 without reducing the initial loss.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the starting parameter and update budget identical to compare rates fairly. Rate one-quarter approaches the optimum from below; rate three-quarters crosses it and comes back.

Their final losses happen to match after two updates, despite different paths. Rate one repeatedly overshoots by the same distance, so using the correct gradient alone does not ensure improvement.

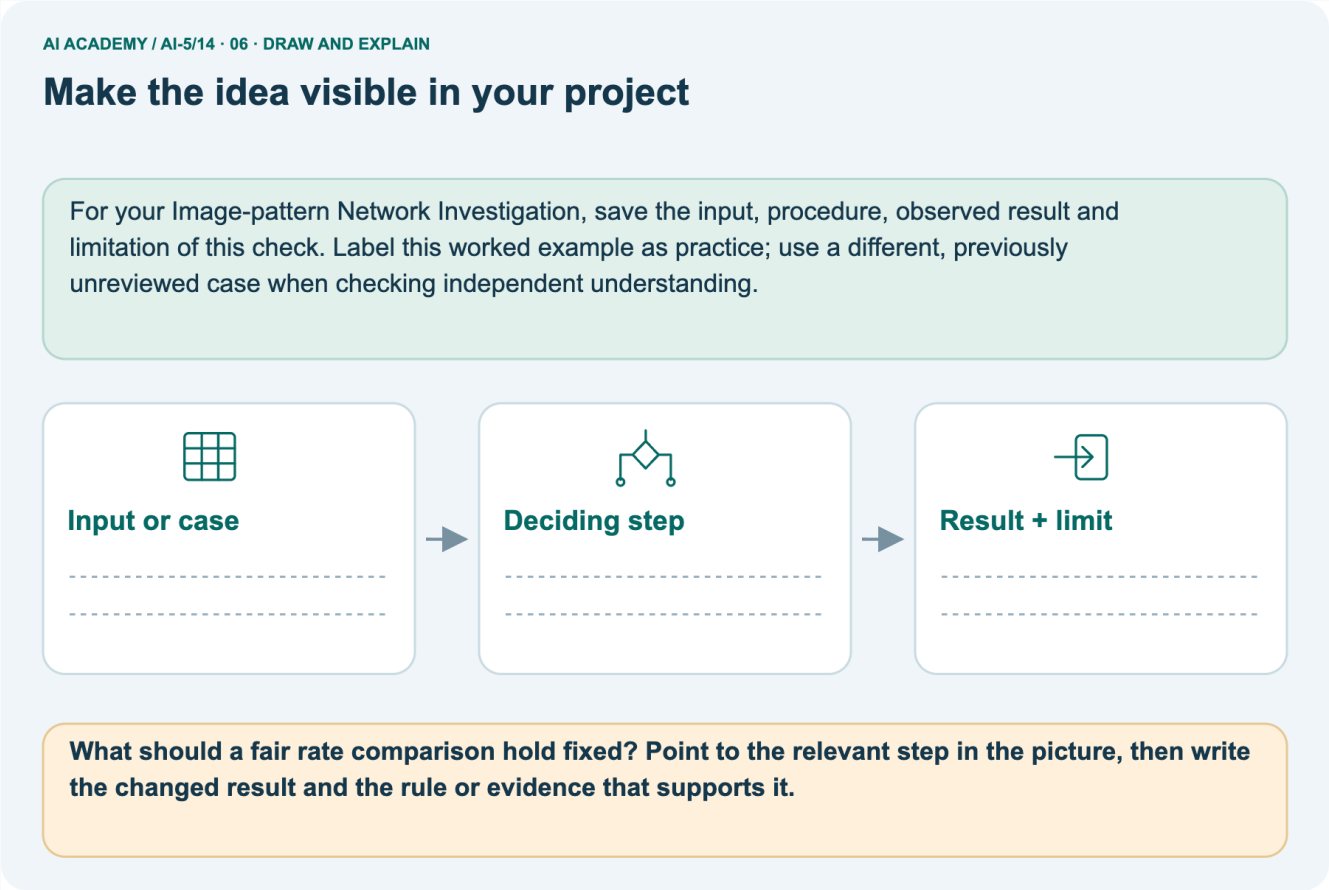
Limit: These quadratic paths explain step size. Real nonconvex networks can have more complicated rate behaviour.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These quadratic paths explain step size. Real nonconvex networks can have more complicated rate behaviour.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Use the same objective, start a = 0 and two-update budget; report full paths, not only the preferred run's endpoint. Changed-case check: Its path is 8 then 0, with loss 16 at both steps.

The update alternates across 4 without reducing the initial loss.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

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▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.